

Mathematics

Quadratic Equations

Given

$$ax^2 + bx + c = 0$$

The roots are

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The discriminant is

$$\Delta = b^2 - 4ac$$

$\Delta > 0$ two real roots

$\Delta = 0$ one repeated root

$\Delta < 0$ two complex roots

Factoring

Difference of squares

$$a^2 - b^2 = (a - b)(a + b)$$

Perfect squares

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

Exponent Rules

$$a^m a^n = a^{m+n}$$

$$\frac{a^m}{a^n} = a^{m-n}$$

$$(a^m)^n = a^{mn}$$

$$(ab)^n = a^n b^n$$

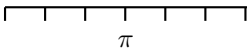
$$a^{-n} = \frac{1}{a^n}$$

Logarithms

$$\log(ab) = \log a + \log b$$

$$\log\left(\frac{a}{b}\right) = \log a - \log b$$

$$\log(a^n) = n \log a$$



“Mathematics is the art of
explaining why.”